

CLO 3 (Engineering Ethics)

Q: Which ethical theory would a mechanical engineer employ to justify their position in the following case? Analyze the given situation with reference to at least **three applicable ethical theories**, support your answer with logical reasoning, and discuss how such problems may be resolved.

A mechanical engineer is tasked with designing and overseeing the construction of a new transportation system, which includes the development of an extensive rail network. This project will pass through a densely populated urban area, causing concerns about displacement of residents and potential environmental pollution. However, the rail network is crucial for reducing traffic congestion and enhancing public transportation efficiency in the city.

Q: How would you approach the following situation using **line drawing or flowcharting** techniques to analyze the ethical implications and decide on the appropriate course of action?

As a mechanical engineer at an aerospace company, you are responsible for overseeing the quality control of aircraft components. During a routine inspection, you discover that a batch of components does not fully meet the stringent quality specifications required for safety-critical parts. The deviations are minor and unlikely to cause immediate issues, but they do not comply with the established standards. The production manager urges you to approve the batch to meet tight delivery schedules and avoid significant financial penalties. He assures you that the components are still safe for use and that rejecting the batch would lead to costly delays and potential layoffs.

Q. Read the following case and answer the questions below:

In a small town with a rich historical past, an old factory site once celebrated for its textiles in the late 19th century is scheduled for redevelopment into a community center and apartments. As an environmental engineer conducting assessments, you discover that the soil and groundwater are contaminated with heavy metals and industrial solvents from the factory's operations. This contamination poses significant health risks to future residents and visitors.

a. Define **whistleblowing** and categorize its **various types** with real-life examples for each type. Explain the ethical considerations typically associated with whistleblowing.

b. Would you consider whistleblowing in this scenario? If so, specify the type of whistleblowing you would pursue and justify your choice using relevant **ethical theories and principles**.

c. Discuss the potential **consequences, ethical implications**, and **stakeholders** involved in your decision to blow the whistle or refrain from doing so.

Q. Which ethical theory would an engineer employ to justify their position in one of the following three cases? Analyze the given situation with reference to the applicable ethical theories and support your answer with logical reasoning and discuss how such problems may be resolved. Choose from the following cases:

1. A civil engineer is working on a large infrastructure project, which involves constructing a highway through a dense forest area. The project has faced opposition from environmentalists who are concerned about the potential destruction of natural habitats and the disruption of ecosystems in the region. However, the highway is essential for improving transportation and economic development in the area.

2. A software engineer is working on a critical project for a medical device company. The software is designed to control an implantable medical device that assists patients with heart conditions. During the testing phase, the engineer discovers a potentially life-threatening bug in the software. Fixing the bug would require significant delays and could result in a halt in the project's progress, causing financial losses to the company and potentially affecting patient care.

3. The World Trade Organization (WTO) was established to oversee trade agreements, enforce trade rules, and settle disputes. Some troublesome issues have arisen when WTO has denied countries the right to impose environmental restrictions on imports from other countries. Thus, for example, the United States may not impose a ban on fish caught with nets that can endanger other sea life such as turtles or dolphins, while European countries and Japan will not be able to ban imports of beef from U.S. herds injected with antibiotics. Yet, other countries ban crops genetically modified to resist certain pests, or products made there from, unless labeled as such.

Q. Analyze and discuss the given scenario with **PEC codes of conduct** and provide possible resolutions.

1. A mechanical engineer is responsible for procuring parts from various suppliers for a construction project. The engineer discovers that some suppliers are providing substandard parts that may compromise the overall quality and safety of the project. Discuss the dilemma and provide resolutions under PEC codes of conduct.

2. A mechanical engineer is working in a manufacturing plant in Pakistan where heavy machinery is used. The engineer notices that some of the safety guards on the machines are missing or damaged, posing a significant risk to the workers' safety. Discuss the dilemma and provide resolutions under PEC codes of conduct.

3. An engineering consultant in Pakistan is involved in the process of allocating a large government project to different engineering firms. The consultant has close personal ties with one of the firms bidding for the project, and they stand to benefit significantly if that firm wins the contract. Discuss the dilemma and provide resolutions under PEC codes of conduct.

Q. For each of the following three cases, identify the **factual, conceptual, and moral** issues, and dilemmas involved. Additionally, explain why you consider the identified aspects to be moral values or dilemmas.

1. An electronic engineer informed his company that, for a relatively minor cost, a kitchen appliance could be made to last several years longer by using a more reliable component. However, the firm decides not to implement the change, citing cost concerns and the strategic benefit of built-in obsolescence, which ensures customers will need to purchase replacements more frequently.

2. Your boss, aware that you have many friends throughout the company, relies on you for feedback about employee morale and other staff-related matters. Recently, she asked you to report any behavior that might violate company policies, such as taking office supplies home or making personal long-distance calls. List the ethical issues you would like to discuss with her before responding to this request.

3. Harris, an electronics engineer, has the authority to influence the selection of suppliers for his company's annual purchases. During Eid, he often receives small tokens from various salespeople, such as pens or diaries. This year, however, one salesman sends him an expensive briefcase monogrammed with his initials. Should Harris accept the gift? Should he take any further action? Justify your response based on ethical reasoning.

Q. Complete the following given chart by providing the required information about the four types of Ethical theories.

Name of the Theory	Definition	Advantages	Disadvantages
1. Utilitarian Theory			
2. Deontology Theory			
3. Rights Theory			
4. Virtue Theory			

CLO 2 (Writing Business Messages)

Principles of Effective Communication

Q. You are the Engineering Manager at a manufacturing company. Using modified block format, write a **letter** to inform a prospective mechanical engineer, Sarah Smith, that she was not selected for the position of Mechanical Engineer. Apply concreteness and courtesy **principles of effective communication** and write the letter with a focus on maintaining a positive relationship and encouraging them to apply for future opportunities within the company.

Q. Rewrite the following communications by consciously targeting the 7 Cs of effective communication. The major Cs violated are indicated in parentheses.

1. Let's meet tomorrow too discuss the product launch event. Plz b on time. (**Correctness**)
2. In regard to our telephonic conversation today, I am emailing you the plan of the project. The deadline is for one week. (**Completeness and Correctness**)
3. We hereby wish to let you know that our company is pleased with the confidence you have reposed in us (**Conciseness**)
4. I don't appreciate how your team ignores requests for collaboration from my team. The work we do is equally as important as your work. Could you make certain that your team collaborates more readily from now on? (**Courtesy**)

Q. Rewrite the following e-mail using consideration and clarity factors.

Dear Samantha,

We will have to meet urgently as our new client wants some changes in the product specifications. We may also need help from the design team.

Best regards,

Meeting Notices & Minutes

Q: Your Engineering Director, Robert Lee, has requested you to write a **meeting notice** for the upcoming Research and Development Strategy Meeting by applying all the necessary **principles of effective communication**. In the notice, inform all team members about the schedule of the strategy meeting. Include

the **agenda** items, such as discussion of R&D priorities, allocation of resources, and exploration of new technologies. Also, write the **minutes** of this assumed meeting.

Q. Your Manager, Sarah Johnson, has requested you to write a **meeting notice** for the upcoming Quality Assurance Audit. In the notice, inform all team members about the schedule of the audit meeting. Include the agenda items, such as review of quality metrics, identification of non-conformances, and discussion of corrective actions. Also, write the **minutes** of this assumed meeting.

Q. Your Manager, Jane Miller, has requested you to write a meeting notice for the upcoming Quarterly Performance Review. In the notice, inform all team members about the schedule of the review meeting. Include the agenda items, such as individual performance assessments, goal setting for the next quarter, and an opportunity for employees to provide feedback and suggestions. Also, **write the minutes** of this assumed meeting.

Informational Report:

Q: You are a Senior Mechanical Engineer at your company. Using memo format, write an **informational report** on the importance of safety standards in mechanical engineering projects. Highlight how adherence to safety protocols impacts project success, risk management, and overall organizational reputation. Address the memo to the Project Management Team. Add a meaningful and descriptive heading for each section of the report.

Q. You are the Lead Mechanical Engineer at a prominent engineering research institute. Using letter format, write an **informative report** to a technology magazine editor, providing information on recent innovations in mechanical engineering. Highlight how these innovations are reshaping industries such as automotive, aerospace, and manufacturing. Address the letter to the Editor-in-Chief of the Technology Magazine. Add a meaningful and descriptive heading for each section of the report.

Q. Using memo format, write an informational report on the importance of effective communication in the business world. Highlight how communication skills impact professional success, team dynamics, and overall organizational performance. Add a meaningful and descriptive heading for each section of the report.

Business Letters:

Q. You are the Director of Human Resources at a financial services firm. Using semi-block format, write a **letter** to inform an employee, Emily Johnson, about changes in the company's operational structure that will require a demotion from her current role. Also, apply the rules of effective communication in your message.

Q. You are the Engineering Manager at a manufacturing company. Using modified block format, write a **letter** to inform a prospective mechanical engineer, Sarah Smith, that she was not selected for the position of Mechanical Engineer. Apply concreteness and courtesy **principles of effective communication** and write the letter with a focus on maintaining a positive relationship and encouraging them to apply for future opportunities within the company.

Q. You are the Customer Service Manager of an online retail store. Write a letter to inform a customer that the product they recently purchased is out of stock, and you are unable to fulfill their order. Use full block format and write an appropriate buffer sentence and underline it (if applicable).

Q. Based on the job advertisement below for the Junior Mechanical Engineer position, write a cover letter as a recent mechanical engineering graduate. Highlight your passion for engineering, proficiency in CAD software, and problem-solving skills. Showcase relevant coursework or projects that demonstrate your suitability for the role. Craft a compelling cover letter that expresses your enthusiasm to contribute to innovative engineering solutions and be a valuable member of the team.

Join our team as a Junior Mechanical Engineer!

Qualifications:

- Bachelor's degree in Mechanical Engineering (recent graduates welcome)
- Solid understanding of mechanical principles and design
- Proficiency in CAD software and engineering simulation tools
- Strong problem-solving and communication skills

Benefits:

- Competitive entry-level compensation
- Supportive work environment for growth
- Exciting engineering projects

How to Apply: Email your resume and cover letter to careers@innovateengineers.com with subject "Junior Mechanical Engineer Application."

Embark on a rewarding engineering career with us!

Application Deadline: August 3, 2023

Q. Based on the job advertisement above for the Junior Mechanical Engineer position, prepare a functional **résumé** as a recent mechanical engineering graduate. Focus on the major components most relevant to this role, such as key skills, coursework, or projects. Briefly explain why you chose to emphasize these components.

Memorandum

Q. You are the Head of Product Development at TechGears, a company specializing in advanced mechanical components. Recently, a competitor has launched a groundbreaking new product that may affect our market share. Write a **memorandum** to the CEO of TechGears, informing them about the competitor's new product and proposing a strategic response to this development. Additionally, ensure your memo incorporates at least **6 special elements** typical of professional memos.

Q. You are the Chief Engineer at GlobalTech Manufacturing, a company specializing in precision engineering solutions. Recently, a competitor has introduced a cutting-edge manufacturing process that could impact your market leadership. Write a **memo** to the Executive Board, informing them about the

competitor's new process and proposing a strategy to enhance GlobalTech's capabilities and maintain its competitive edge. Incorporate at least **6 specialized elements** that are peculiar to the memo format.

Q. In the recent budget, there was a significant reduction of duties on imported cars. You are the Sales Manager of *Speed Motors*, a local automobile company. Due to the reduction in import duties, it is feared that the car sales of local companies will drop. Write a memorandum to the Managing Director of *Speed Motors*, informing him about the impact of the new development.

Miscellaneous

Q. Differentiate the following with at least one real-life example for each:

- High Context Culture vs. Low Context Culture
- Verbatim Minutes vs. Minutes of Narration
- Lateral Flow of Communication vs. Grapevine Communication
- Feminine Culture vs. Masculine Culture
- Minutes of Resolution vs. Verbatim Minutes
- Annual General Meeting vs Departmental Meeting
- Statuary Meetings vs Working Parties' Meeting
- Board Meetings vs. Management Meetings